

EDUCATION

University of California Irvine | *Irvine, CA*

Sep 2025 - Present

Electrical Engineering Bachelors of Science in the Henry Samueli School of Engineering

Spring Valley High School | *Columbia, SC*

Aug 2021 - Jun 2025

High School Degree in a Discovery (Research Magnet Program) | GPA: 3.91/4.0

PROJECTS

Systems Subteam CubeSat | *UCI CubeSat*

Aug 2026 - Present

- Completion CubeSat satellite cycle, ensuring all subsystems integrated successfully to meet overall mission requirements.
- Managed integration across avionics, communications, and power teams while strictly tracking mass, power, and link budgets.
- Developed core technical documentation, including Concept of Operations (CONOPS) diagrams and Interface Control Documents (ICDs).
- Validated orbital paths via Systems Tool Kit (STK) and executed hardware-software verification on a FlatSat test bench.

Oscilloscope (GitHub Repository)

Aug 2026 - Sept 2026

- Designing a general-purpose oscilloscope from scratch to explore how analog front-end design affects a microcontroller's usable ADC resolution.
- Compares three architectures on one board: direct ADC sampling, hardware auto-ranging, and an external 16-bit ADC.
- The auto-ranging stage level-shifts the input to 0-3.3V, then uses window comparators to detect voltage bands and gain stages to fill the ADC's full range.
- Custom KiCad PCB, hand-soldered, with a Python app for real-time waveform visualization.

AI Empowered Chess Game ([GitHub Repository](#))

Mar 2026 - Apr 2026

- Designed and implemented a fully playable modified chess game in Python (Pygame), replacing traditional checkmate logic with a custom point-based victory system where the first player to reach 15 points wins.
- Reworked core chess mechanics so each captured piece contributes a predefined point value, shifting strategy toward material optimization and risk-reward decision-making with a real-time side-panel scoreboard.
- Currently training a Reinforcement Learning agent (Q-learning/policy-based methods) to learn the modified ruleset from scratch and compare AI adaptation against human player behavior.

RESEARCH

Research on Investigating microbes on combating pathogenic fungi | *Discovery Magnet Program, SC*

Aug 2023 - May 2024

- Designed and executed a quantitative biological degradation study using common bacteria at multiple concentrations to evaluate its effectiveness against a pathogenic fungus.
- Developed a MATLAB-based image processing pipeline that converted before-and-after plate images to 8-bit grayscale and quantified fungal degradation using pixel intensity analysis across 120+ samples. Then applied statistical analysis, including.
- Presented findings at the 2024 South Carolina Junior Academy of Science Conference and 2024 University of South Carolina Region II Science Fair.

Research on Purifying water of Bacterial Infection using fungal colonies | *Discovery Magnet Program, SC*

Aug 2022 - May 2023

- Designed and conducted a controlled bioremediation study evaluating the effectiveness of a fungi in reducing bacterial contamination in water across multiple substrates (coffee grounds, sawdust, wheat straw). Tested over 120 trials.
- Authored a comprehensive research paper and presented findings at the 2023 South Carolina Junior Academy of Science Conference and 2023 University of South Carolina Region II Science Fair.

AWARDS

Honorable Mention at the 2024 University of South Carlina Region II Science Fair for Research on Investigating microbes on combating pathogenic fungi

2023 Stockholm Junior Water Prize from The Water Environment Federation for Research on Purifying water of Bacterial Infection using fungal colonies

Third Place at the 2023 University of South Carlina Region II Science Fair for Research on Purifying water of Bacterial Infection using fungal colonies

Skills

Programming (C/C++, Python, Java, Verilog)	Data Analysis (MATLAB, Matplotlib, ImageJ)	Microcontroller (Arduino)	Embedded Systems & Circuit Design	CAD & Mechanical Design
Machine Learning (Reinforcement Learning)	Microbiology	PCB Design (KiCad & Altium)	Autodesk Fusion	System Integration